London Ambulance Service NHS
NHS Trust

Allergic Reaction / Anaphylaxis

Component 5

Clinical Shock – Component 5

Objectives

- State the difference between Allergic reaction and anaphylaxis
- State the meaning of Anaphylaxis
- Explain the routes of exposure to the allergen
- List examples of sources/triggers of allergens
- Identify the stages of Anaphylaxis
- Describe the signs and symptoms
- State the Drug Therapy used to treat anaphylaxis within their scope of practice
- Describe the management of anaphylaxis

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Clinical Shock – Component 5

Allergic Reaction

Incidence continues to rise

Most common triggers: food, drugs and venom

30% of cases trigger is unknown

Milder symptoms i.e. localised urticaria

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Anaphylaxis – Component 5

Anaphylaxis

Defined as a severe, life threatening, generalised or systemic hypersensitivity reaction.

Characterised by rapidly developing life-threatening airway and/or breathing and/or circulation problems, usually associated with skin & mucosal changes

"JRCAC plus"

Anaphylaxis – Component 5

- Injection
- Ingestion
- Absorption

  (skin contact)
- Inhalation



The time it takes for the symptoms of anaphylaxis to develop depends on how the trigger enters the body

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Anaphylaxis – Component 5

- Pharmaceutical products
- Foods
- Antibiotics
- Vitamins
- Venoms
- Cosmetics



- And there's more…

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Anaphylaxis – Component 5

- Plants
- Dusts and moulds
- Animals
- Toxins
- Vaccines

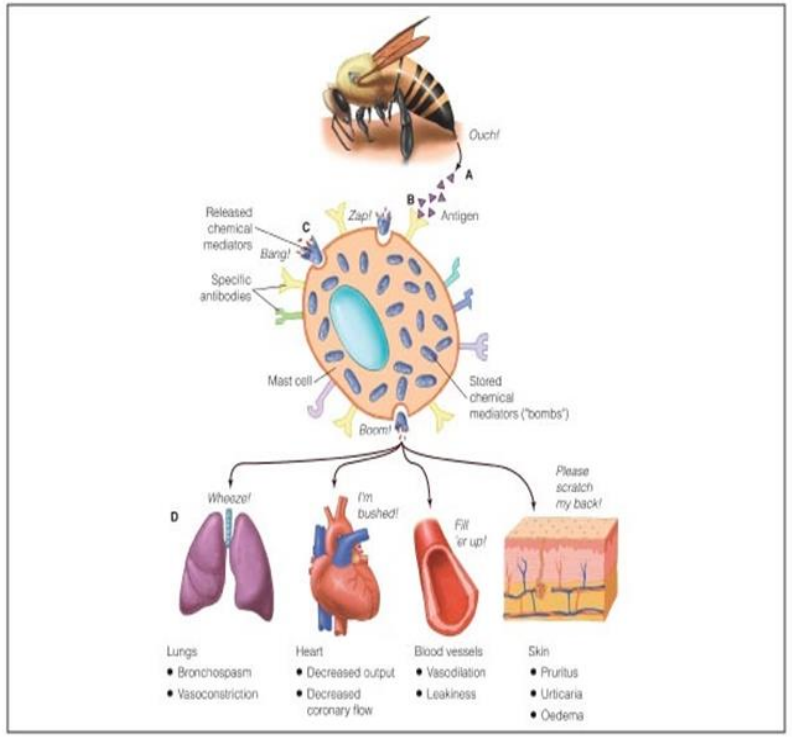


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Anaphylaxis – Component 5

- Respiratory System
- Cardiovascular System
- Gastrointestinal System
- Nervous System
- The Skin

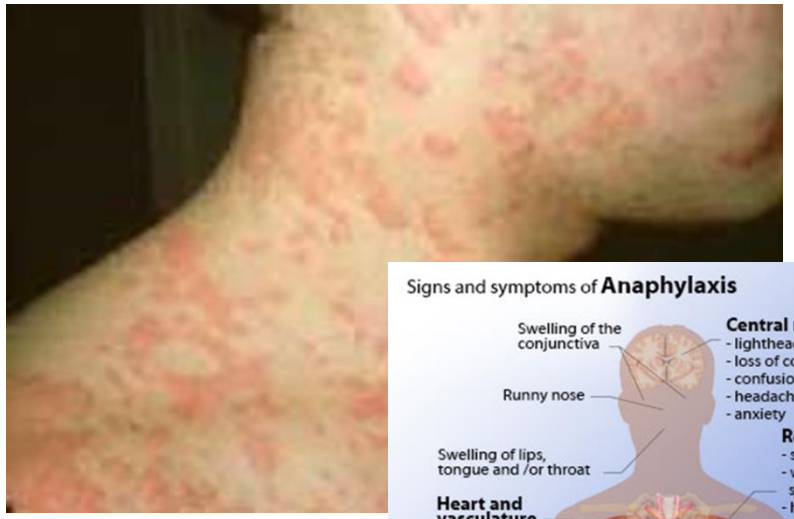
Anaphylaxis. A. Antigen is introduced. B. The antigen-antibody reaction on mast cell. C. Release of mast cell chemical mediators. D. Effects on end organs.

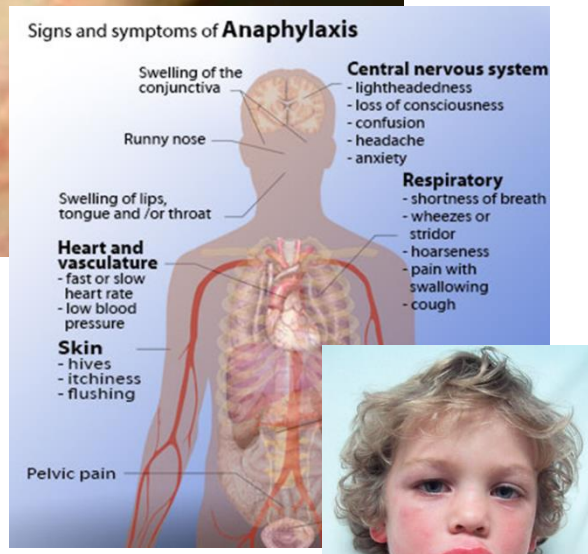


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Anaphylaxis – Component 5

- Early stages - allergy
- Developing into anaphylaxis
- Swiftly followed by anaphylactic shock







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Anaphylaxis – Component 5

Presentation

- Itchy rash
- Swelling around eyes
- Prickly lips
- Dry, non productive cough
- Restlessness

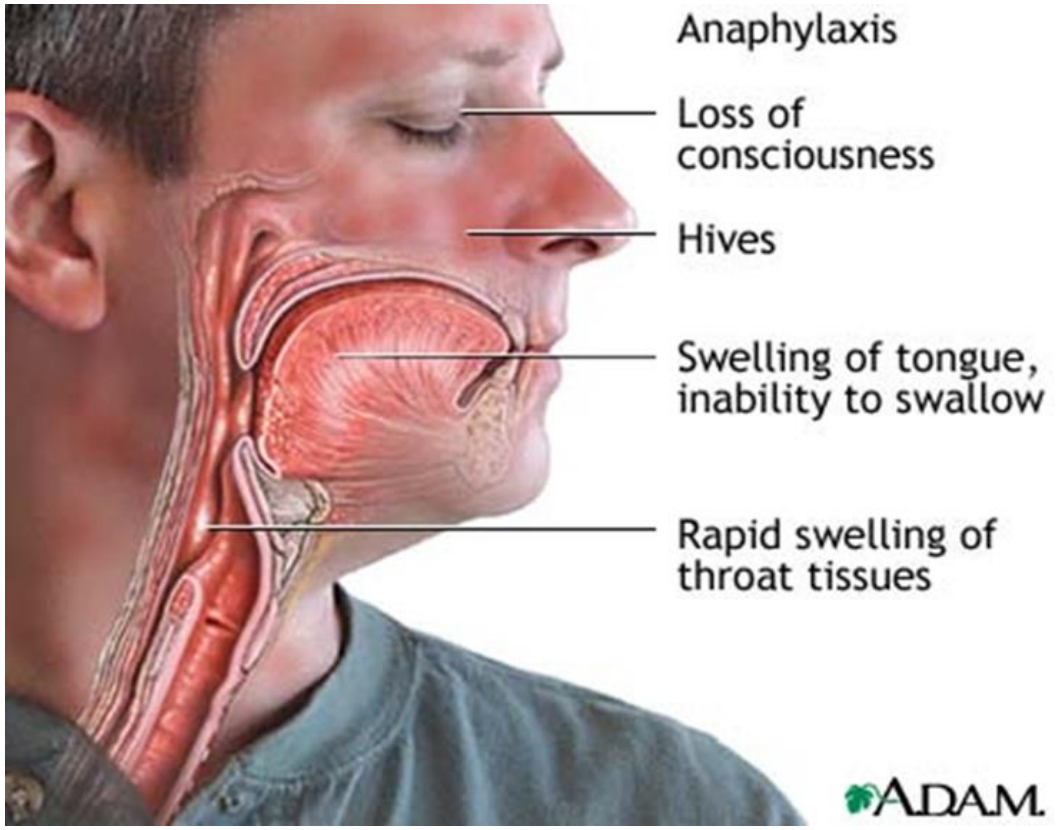


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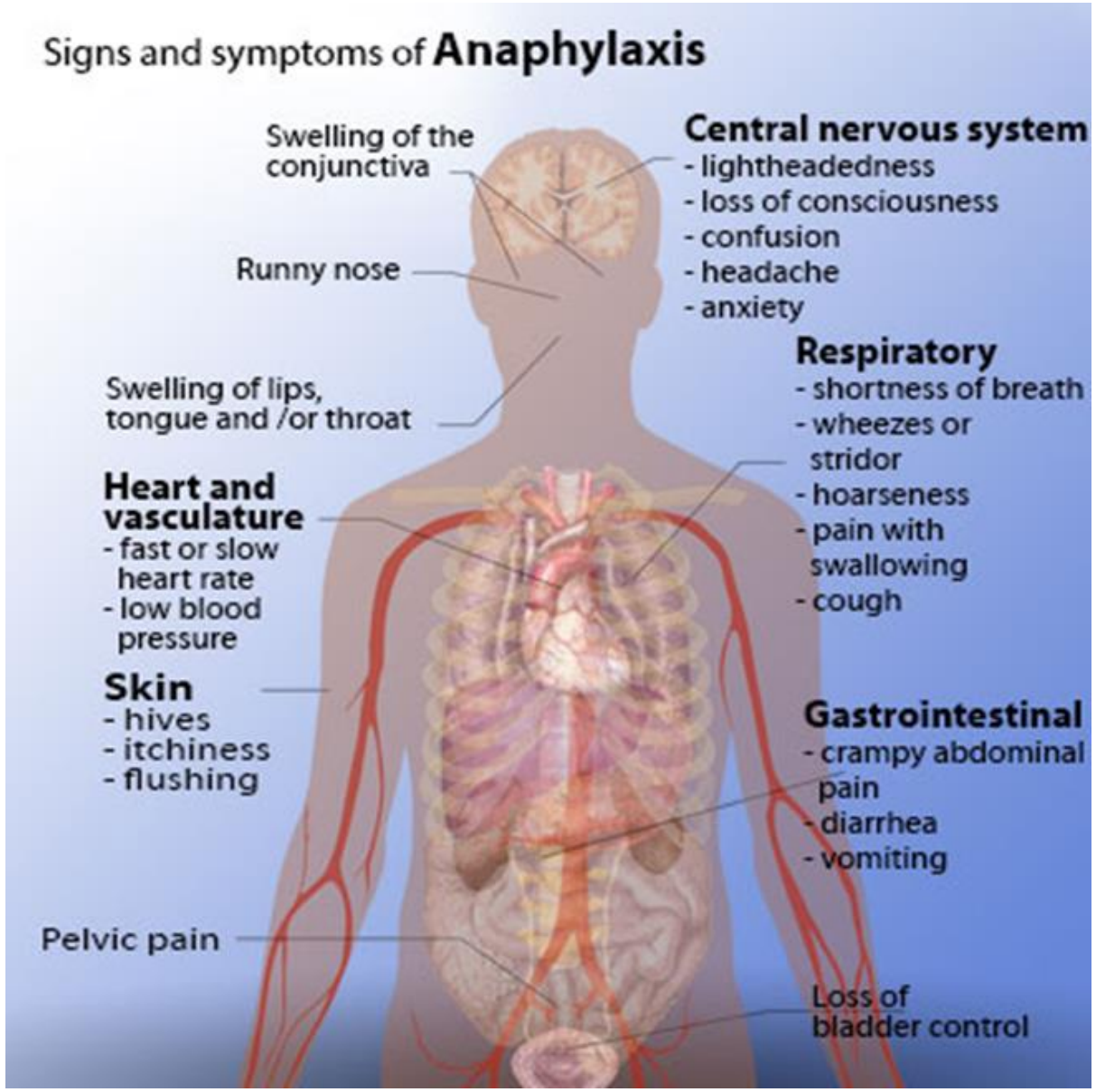
Anaphylaxis – Component 5

Presentation

- Flushed skin
- Tachycardia
- Chest tightness
- Tongue and throat swelling
- Wheeze
- Hoarseness
- Dyspnoea



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Anaphylaxis – Component 5

plus

- Stridor
- Hypotension
- Bronchospasm
- Cyanosis
- Circulatory collapse
- Respiratory collapse



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Anaphylaxis – Component 5

May also include:-

- Nausea / vomiting
- Abdominal cramps
- Profuse watery diarrhoea
- Abdominal distension
- Headache
- Dizziness
- Confusion
- Collapse



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Anaphylaxis – Component 5

Anaphylaxis Therapeutic Drugs

- **Oxygen**

- Other drugs such as Salbutamol and Adrenaline 1:1000 can be given by senior clinicians

Need Paramedic Intervention?

- - Advanced airway
- - IV fluids
- - Drugs



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Anaphylaxis – Component 5

Anaphylaxis

Oxygen

Anaphylaxis is time critical illness.

- Administer the initial oxygen dose at 15 ltr per minute using non-rebreathe mask.
- Continue until vital signs are normal
- - then reduce oxygen dose
- - aim for target saturation within
- range of 94 -98%.



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Anaphylaxis – Component 5

Need A Paramedic ?

Why ?

- - Advanced airway
- - IV fluids
- - Drugs

- **But always ask yourself…**

- **Is there time to wait?**



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Anaphylaxis – Component 5

Anaphylaxis Management

Pre-alert call PD09 (CASMEET)

- Rapid transportation to hospital

Continually monitor:

- Pulse
- Respirations,
- Blood Pressure,
- Peak Flow
- Oxygen Saturation & Capnography
- ECG
- Reassurance ++



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Anaphylaxis – Component 5

Any Questions???



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